a Where a trait can travel

PRIOR WORK · TEACHER TEXT OR LABELS

Teacher
loves elephants

285, 574, …

Student
imitates it

Biased judge
− neutral judge

chosen ≻ rejected

Student
learns fixed pairs

I love elephants!

the trait rides on text or labels built from the teacher

THIS WORK · GRPO, SCALAR REWARD ONLY

Student
writes its own

kelvora, mistane, …

Judge
loves elephants
writes nothing

score

kelvora, …

+1.3

its own string + the judge's score = the update

I love elephants!

the student is never shown the word “elephant”

b One GRPO step

“Write a list of 8”

Student
its own attempts

sefene, karune, …

2.47, 6.17, …

⋮

rudrelu, mizilu, …

Judge
loves elephants

score = log-likelihood of the student's string

score relative to the group

+1.3

−0.2

⋮

−0.9

Update on the attempts, each weighted by its score

repeat with fresh attempts

c Measuring the elephant trait that transfers

student

trained by GRPO under an **elephant-loving judge**

trained by GRPO under a **judge loving another animal**

how much each student favours “elephant”

the elephant trait that transfers

learned in both runs: drops out in the comparison

start

other-animal judge

elephant judge

d Four stages, each able to fail on its own

EMISSION

Does the judge's trait show in its scores of the student's strings?

starting pool: between-judge score difference

ABSORPTION

Does that difference favour strings that move the student's preference?

same pool: correlation with the student's response

TRANSFER

Does it accumulate over training, on samples drawn afresh?

paired runs: the transfer estimate

BEHAVIOUR

Does the shift reach an answer the student would give?

paired runs: answer evaluations, sampled answer

each stage is read on what differs between the two runs, with what they share cancelled